AIHM Task 7 Data Collection Report

Data collected by Lyan and Nofe | Galaxy smartwatch heart rate exports | 29 October 2026

The purpose of this report is to document two heart rate datasets gathered in accordance with the requirements of AIHM Task 7 by Lyan and Nofe. Lyan and Nofe have provided separate exports. The team would like to state that the data was collected using the watch application created in Task 2 (T2). The names Lyan and Nofe are mentioned to indicate the collectors of the data in this report and they do not refer to the persons who were being tested.

Data collection and source files

Lyan exported her heart rate data into a CSV file, while Nofe exported his into a ZIP file using Samsung Health and included additional JSON data. Both sets of data were processed independently. The available data includes timestamps and heart rate values, with the devices being identified by aliases. The collector labels are simply there to differentiate between the exports and do not refer to any wearer identification..

Preparation procedure

1. Read the heart rate records from each collector's export, parsed measurement times and values, and retained technically valid readings.
2. Separated spot measurements from interval data. Nofe's linked JSON produced minute level heart rate bins. Lyan's 744 interval summary records without corresponding detail JSON stayed in the source record table and were not converted into invented minute values.
3. Grouped the retained measurements into five minute windows for analysis while keeping the individual readings in a separate table.
4. Replaced raw watch device UUID values with device aliases and kept an audit of the number of source and resulting records.

Prepared data summary

Measure	Lyan collected export	Nofe collected export
Heart rate source records	6,207	974
Retained heart rate measurements	5,463	9,757
Type of retained measurements	5,463 spot readings	766 spot readings and 8,991 minute bins
Populated five minute windows	5,461	2,687
Dates with at least one retained reading	128	39
Range of retained readings in UTC	4 Nov 2025 to 6 Oct 2026	16 Dec 2024 to 6 Oct 2026
Longest run of dates with readings	28 Aug to 13 Sep 2026: 17 dates	7 Oct to 24 Oct 2025: 18 dates

A date with readings means that at least one retained measurement occurs on that UTC date. The first and last dates show the range covered by each export; they are not a count of continuously recorded days.

Dataset organization

Output	Description

heart_rate_measurements.csv	One row per retained heart rate observation, with reading time, heart rate, observation type and source and device aliases.
heart_rate_5min.csv	Five minute windows populated by the available readings; includes mean heart rate and the number of observations in each window.
heart_rate_source_records.csv	Source level audit table showing the original heart rate records, including interval summaries.
audit.json	Processing counts and settings for the corresponding export.

The prepared files preserve the two collected exports separately. The export provided by Nofe contains a Galaxy Watch7 device profile with model SM-L310. The input provided by Lyan was a heart rate CSV. The individual readings and five minute summaries are the prepared Task 7 data outputs.